

A moderate increase in daily protein intake causing an enhanced endogenous insulin secretion does not alter circulating levels or urinary excretion...

To study the effect of a moderate increase in insulin secretion produced by an increased daily protein intake on dehydroepiandrosterone sulfate (DHEAS), a balanced randomized crossover trial consisting of three strictly controlled dietary regimens was performed in six healthy male volunteers. The basic diet (B) contained 50 g protein/d; diets P and M (also basic diets) were enriched with either 32 g protein/d (P) or 10 mmol L-methionine/d (M). Methionine was given (as a specific nonprotein source of endogenously derived sulfate) to control for possible confounding effects on DHEAS due to an increased sulfate supply. At the end of each 4-day diet period, blood and 24-hour urine samples were collected. Fasting plasma levels of testosterone, cortisol, insulin-like growth factor-I (IGF-I), and insulin, as well as urinary output of total (hot acid-cleaved) testosterone conjugates and 3 α -androstenediol glucuronide, did not show significant changes in response to dietary manipulations. Endogenous sulfate availability (as reflected by renal sulfate output per 24 hours) approximately doubled with diets P and M. However, plasma levels (6.3 $\pm$ 1.5, 6.8 $\pm$ 1.8, and 6.9 $\pm$ 2.1 micromol/L for B, P, and M, respectively) and urinary excretion (8.8 $\pm$ 9.8, 9.4 $\pm$ 11.2, 8.0 $\pm$ 8.3 micromol/d) of DHEAS remained unaffected. Considering the clear increments ($P < .01$) in urinary C-peptide excretion with diet P (20.4 $\pm$ 10.3 nmol/d) versus diets B and M (12.6 $\pm$ 5.1 and 13.2 $\pm$ 3.6 nmol/d), respectively, our results suggest that a moderately strong diet-induced increase in daily insulin secretion does not alter urinary and plasma levels of DHEAS.